**ECA18 – OPTOELECTRONICS AND OPTICAL COMMUNICATION**

**CO1 ENGINEERING DESIGN ASSIGNMENT**

**Standards-Compliant Optical Fiber Link Design: From Communication Requirement to Deployment Decision**

**Course Outcome – CO1**

Apply the concepts of guided modes in step-index and graded-index fibers, evaluate losses (Absorption, Scattering, Bending), dispersion mechanisms, and compensation techniques, including OSNR.

Nature of Assessment: Open-Ended Engineering Design Assignment

Duration: 1 Week

Maximum Marks: 100

Primary Bloom's Levels: K3 – Apply | K4 – Analyze | K5 – Evaluate | K6 – Create/Design

**THE ENGINEERING DESIGN PROBLEM**

A communication service provider plans to establish a 60 km optical fiber link operating at 1550 nm to carry 10 Gb/s traffic between a central office and a business building, with provision for a later upgrade to 100 Gb/s. The link must use standard ITU-T fiber categories, tolerate tight bends at the building entry, and maintain pulse integrity, polarization performance and signal quality. The design engineer must determine whether the proposed link configuration is technically suitable before deployment.

The proposed design must satisfy the following information and constraints:

| **Parameter** | **Requirement / Given Condition** |
|---|---|
| Transmission distance | 60 km |
| Operating wavelength | 1550 nm |
| Candidate fiber A (step-index) | $n$ = 1.4500, $n$ = 1.4450, core radius a = 4.0 μm |
| Candidate fiber B (step-index) | $n$ = 1.480, $n$ = 1.465, core radius a = 25 μm |
| Fiber losses | Absorption = 0.04 dB/km; Rayleigh scattering = 0.11 dB/km; Bending (distributed) = 0.03 dB/km |
| Transmitter power | +3 dBm |
| Connectors and splices | Four connectors, 0.4 dB each; ten splices, 0.08 dB each |
| Building-entry bend loss | 0.8 dB with G.652.D-type fiber; 0.1 dB with G.657.A-type fiber |

| | |
|---|---|
| Receiver requirement | Sensitivity = −20 dBm; required system margin = 6 dB |
| Chromatic dispersion | G.652.D: D = 17 ps/(nm·km); G.655: D = 4.5 ps/(nm·km); source spectral width $\Delta\lambda$ = 0.4 nm; input pulse width 30 ps; bit period $T_b$ = 100 ps (10 Gb/s) |
| Dispersion-compensating fiber | $D_{DCF}$ = −100 ps/(nm·km); attenuation = 0.5 dB/km |
| PMD and birefringence | PMD coefficient = 0.1 ps/√km; PMD limit = 0.1$T_b$; B = 1 × 10 (standard fiber), B = 4 × 10 (PM fiber); beat length $L_B = \lambda/B$ |
| OSNR study (0.1 nm reference bandwidth) | Signal power = −16 dBm; ASE noise power = −39 dBm; required OSNR = 17 dB |
| Optical amplifier | Signal gain = 14 dB; ASE noise increase = 16 dB |

**You are not given:**

· the final fiber type, ITU-T category and propagation choice;

· the mode-selection decision;

· the dominant loss mechanism requiring attention;

· whether dispersion compensation is required and, if so, its length, and whether PMD is acceptable;

· whether the proposed amplification strategy is acceptable from an OSNR perspective; or

· the final deployment recommendation.

You must make these decisions from the communication requirement, calculations, comparison of alternatives and engineering reasoning.

Your objective is not merely to calculate individual parameters. Your objective is to develop a technically justified optical-link configuration and defend the final engineering decision.

**STAGE 1 – INTERPRET THE COMMUNICATION REQUIREMENT**

10 Marks | K3/K4 – Apply and Analyze

Work backward from the required 60 km, 10 Gb/s communication service (with upgrade provision) before beginning detailed calculations.

Determine and discuss:

· the purpose of the 1550 nm operating condition and the nature of light (ray versus wave behaviour, coherence, polarization) that governs propagation;

· the required light-acceptance and guided-mode characteristics (NA, V-number, LP modes, cut-off) of the candidate fibers;

· the propagation characteristics and ITU-T fiber categories (G.652.D, G.655, G.657) relevant to a 60 km link with tight building-entry bends;

· the loss and power-budget quantities (absorption, Rayleigh scattering, bending, connectors, splices) that must be controlled;

· the dispersion-related quantities (modal, chromatic, PMD) that determine pulse integrity and the ways of compensating them;

· the signal-quality quantity (OSNR) that must be checked in the presence of ASE noise and its relation to BER;

· the practical constraints (bend radius, upgrade path, system margin) that influence the final fiber-link configuration.

Based on this analysis, identify the characteristics that an acceptable optical link solution should possess.

**Stage 1 Deliverable**

Conclude this stage with a concise Design Requirement Statement defining what the proposed 60 km optical link must achieve.

**STAGE 2 – INVESTIGATE AND SELECT THE FIBER / PROPAGATION CONFIGURATION**

15 Marks | K4/K5 – Analyze and Evaluate

The link must collect sufficient optical power from the source, provide an appropriate modal condition and use a suitable ITU-T fiber category.

Determine the parameters required to make this decision:

· Identify the fiber parameter that measures light-acceptance capability and evaluate the numerical aperture of fibers A and B.

· Identify the parameter that combines wavelength, core size and index difference to predict modal behaviour and evaluate the V-number of both fibers at 1550 nm.

· Select an appropriate approximation for the number of guided modes (step-index and graded-index) and determine the approximate mode count of fiber B; state the cut-off condition and cut-off wavelength that classify fiber A as single-mode or multimode.

· Explain when ray theory is adequate and when wave optics (Maxwell wave equation and LP-mode analysis) is required, using the calculated V-numbers.

· Compare single-mode, step-index multimode and graded-index multimode propagation in terms of modal dispersion and bandwidth-distance performance, and relate G.652.D, G.655 and G.657 fibers to the application.

DESIGN DECISION 1 – CREATE THE PROPOSED FIBER CONFIGURATION

Select the propagation/fiber configuration (including ITU-T category) you consider most appropriate for the intended 60 km high-speed application.

Complete the statement:

I recommend _____ propagation/fiber configuration because _____.

Support your decision quantitatively and technically. The recommendation must be based on modal behaviour, propagation characteristics, ITU-T fiber category and application requirements.

**STAGE 3 – DEVELOP AND EVALUATE THE OPTICAL POWER BUDGET**

15 Marks | K4/K5 – Analyze and Evaluate

The selected link must deliver adequate optical power to the receiver with the required margin. Work backward from the receiver requirement to determine the losses that must be controlled.

You must:

· Identify the single attenuation coefficient that represents the combined effect of absorption, Rayleigh scattering and bending losses.

· Calculate the total fiber attenuation coefficient and the loss introduced by the 60 km fiber.

· Include the connector, splice and building-entry bend losses for both G.652.D-type and G.657.A-type fiber in the practical link budget.

· Determine the total link loss, the received optical power in dBm and the margin against the receiver sensitivity and the required 6 dB system margin.

· Identify the dominant specified fiber-loss mechanism from the calculated values.

· Recommend two practical actions for reducing avoidable link loss.

DESIGN DECISION 2 – CREATE THE PROPOSED POWER-BUDGET CONFIGURATION

Decide whether the proposed power budget is suitable for deployment, whether bend-insensitive (G.657) fiber should be used, and identify the loss mechanism that should receive the greatest attention during optimization.

Complete the statement:

I recommend _____ power-budget configuration because _____.

Support the recommendation using the calculated attenuation, total loss, received power, system margin and dominant loss mechanism.

**STAGE 4 – INVESTIGATE AND SELECT THE DISPERSION / POLARIZATION-MANAGEMENT CONFIGURATION**

15 Marks | K4/K5 – Analyze and Evaluate

The 60 km link must maintain pulse integrity at 10 Gb/s and remain upgradable. The design engineer must determine whether uncompensated chromatic dispersion and PMD are acceptable and whether a dispersion-compensating section or polarization-related measures are required.

Determine and discuss:

· Identify why modal dispersion is absent in single-mode fiber A but dominant in multimode fiber B, and how a graded-index profile reduces it.

· Calculate the chromatic-dispersion broadening $\Delta\tau = D \cdot \Delta\lambda \cdot L$ for G.652.D ($D = 17$ ps/(nm·km)) and G.655 ($D = 4.5$ ps/(nm·km)) with $\Delta\lambda = 0.4$ nm and $L = 60$ km.

· Combine the 30 ps input pulse width with each broadening using an appropriate RSS relationship and compare the result with the 100 ps bit period; analyze the implication for pulse overlap and inter-symbol interference.

· Identify the condition for complete first-order dispersion compensation, determine the required DCF length for the G.652.D link using $D_{DCF} = -100$ ps/(nm·km), and calculate the added DCF loss and the resulting received power and margin.

· Calculate the link PMD $= 0.1 \cdot \sqrt{L}$, compare it with the limit $0.1T_b$ for 10 Gb/s and for a 100 Gb/s upgrade ($T_b \approx 10$ ps), and state one PMD-mitigation technique.

· Identify birefringence and calculate the beat length $L_B = \lambda/B$ for standard fiber and PM fiber; explain the role of PM fiber and why it is or is not needed for this link.

DESIGN DECISION 3 – CREATE THE DISPERSION / PMD-MANAGEMENT CONFIGURATION

Decide which fiber category (G.652.D or G.655) should be used, whether a dispersion-compensation section should be included in the proposed link, and how PMD will be managed.

Complete the statement:

I recommend _____ dispersion/PMD configuration because _____.

Support the decision quantitatively using the pulse-broadening results, calculated compensation length, DCF loss and PMD results.

**STAGE 5 – EVALUATE OSNR AND THE AMPLIFICATION STRATEGY**

10 Marks | K4/K5 – Analyze and Evaluate

The final link must maintain acceptable signal quality in the presence of ASE noise. The amplifier must therefore be evaluated for signal-quality improvement, not merely signal-power increase.

You must:

· Identify the performance metric that should be used to judge received optical signal quality relative to ASE noise.

· Calculate the initial OSNR from −16 dBm signal power and −39 dBm ASE noise power, and state how OSNR is measured (reference bandwidth).

· Determine the signal and noise levels after 14 dB signal gain and 16 dB ASE-noise increase.

· Calculate the resulting OSNR, compare it with the original OSNR and with the required 17 dB, and state the remaining margin.

· Analyze how inadequate OSNR would affect signal quality, Q-factor, BER and system margin.

· Recommend two practical measures for maintaining or improving OSNR in a long-distance optical link.

DESIGN DECISION 4 – CREATE THE OSNR / AMPLIFICATION STRATEGY

Decide whether the proposed amplification strategy should be accepted for the intended application.

Complete the statement:

I recommend _____ amplification/OSNR strategy because _____.

Support the decision using the calculated OSNR values and the expected effect on communication performance.

**STAGE 6 – VERIFY AND CRITICALLY EVALUATE THE COMPLETE LINK**

5 Marks | K5 – Evaluate

Prepare a final verification table using the results obtained from Stages 2–5.

| Requirement | Specified / Target | Achieved | Satisfactory? |
|---|---|---|---|
| Modal condition / fiber category | | | |
| Fiber loss / link loss | | | |
| Received optical power and margin | | | |
| Pulse broadening / DCF requirement | | | |

| | | | |
|---|---|---|---|
| PMD / polarization performance | | | |
| OSNR | | | |

Your verification must explain whether the calculated values are consistent with the original communication requirement.

Create the final configuration rather than merely selecting an existing option. Explain how the individual design choices work together to satisfy the communication requirement.

• a final parameter/design table showing the values adopted in the proposed configuration (fiber category, DCF length, amplifier setting);

• a clearly labelled block diagram of the proposed 60 km link;

• selected OSNR/amplification strategy;

• selected dispersion-compensation and PMD-management arrangement, including the required DCF length;

• calculated optical power budget and received-power estimate;

• selected fiber/propagation configuration, its ITU-T category and its justification;

The final design must include:

Using the results and decisions from Stages 1–6, create a complete 60 km optical fiber-link configuration for the stated application. Your design must integrate the selected fiber/propagation approach and ITU-T category, optical power budget, loss-management strategy, dispersion/PMD-management configuration, and OSNR/amplification strategy.

20 Marks | K6 – Create/Design

**FINAL SYSTEM DESIGN – K6 CREATE / DESIGN**

**FINAL ENGINEERING DECISION (10 Marks | K5/K6 – Evaluate and Create)**

WOULD YOU DEPLOY THE DESIGN YOU CREATED?

State clearly:

YES / NO – WITH TECHNICAL JUSTIFICATION

Defend your decision using evidence obtained from the complete design.

Your conclusion must consider:

- fiber/modal suitability and ITU-T category;
- propagation and bandwidth-distance performance;
- total optical loss and received power;
- dominant loss mechanism and practical optimization;

- chromatic dispersion, PMD and the compensation requirement;
- OSNR and amplification strategy;
- overall suitability for reliable 10 Gb/s, 60 km operation with a 100 Gb/s upgrade path.

**If your answer is NO**

Propose a redesign. Identify exactly what you would change, such as:

- fiber/propagation configuration;
- link-loss management;
- dispersion-compensation configuration;
- amplifier/OSNR strategy;
- launch power or passive-link components.

Explain why you would make the change and predict its effect on link performance.

**SUBMISSION REQUIREMENTS**

Submit one Engineering Design Report.

Your report should demonstrate the complete engineering design process:

REQUIREMENT → ALTERNATIVES → ANALYSIS → SELECTION → LINK DESIGN → VERIFICATION → DEPLOYMENT DECISION

The submission must contain:

- design calculations;
- comparison of fiber/propagation alternatives;
- power-budget calculations;
- dispersion and compensation analysis;
- OSNR evaluation;
- justification of major design decisions;
- final verification table;
- final engineering recommendation.

Calculation results without corresponding engineering reasoning will not receive full marks.

**ASSESSMENT RUBRIC**

| Assessment Criterion | Marks | Primary BL |
|---|---|---|

| | | |
|---|---|---|
| Interpretation of communication/design requirement | 10 | K3/K4 |
| Analysis of fiber / propagation alternatives | 15 | K4 |
| Evaluation and selection of power-budget configuration | 15 | K4/K5 |
| Evaluation and selection of dispersion/PMD-compensation design | 15 | K4/K5 |
| Evaluation and selection of OSNR / amplification strategy | 10 | K5 |
| Creation of the integrated optical-link design | 20 | K6 |
| Verification and final engineering/deployment decision | 15 | K5/K6 |
| TOTAL | 100 | |

**PERFORMANCE EXPECTATIONS**

Excellent

The student systematically evaluates the communication requirement, compares alternatives, performs accurate calculations, makes technically defensible design decisions, verifies the complete link and provides a coherent deployment recommendation.

Good

The student demonstrates correct design methodology and appropriate engineering decisions, with minor errors or limited justification in some areas.

Satisfactory

The student obtains a workable optical-link configuration but relies mainly on standard calculations with limited comparison, justification or critical evaluation.

Needs Improvement

The student primarily reproduces calculations without demonstrating appropriate engineering decisions or connecting the results to the communication requirement.

Create/Design – K6

The student creates an integrated optical-link configuration from the stated communication requirement, combines multiple calculated design elements coherently, presents a technically justified configuration, and defends the final deployment decision using quantitative evidence.